

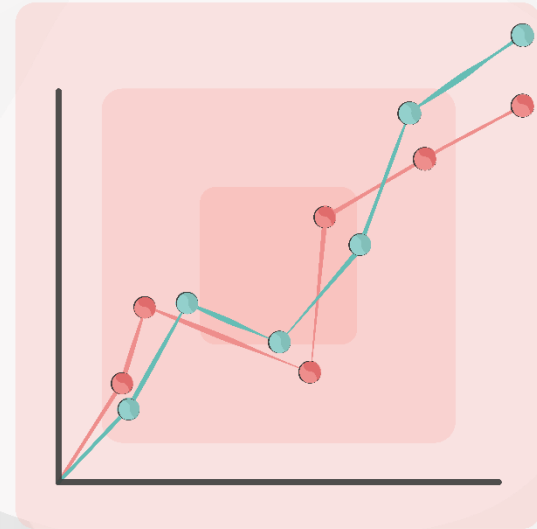


Data engineering

Module: Data pipelines

Topic: On-premise resource management

**Welcome to today's
webinar.**



Ice breaker

Discussion...

- What's the most chaotic tech issue you've ever experienced at work or school?"
- True or False: On-premise systems are outdated and irrelevant in today's cloud-first world?
- What is one key skill or insight you hope to gain from today's session?

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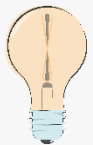
**Submit your responses to the
chat or turn on your
microphone**



Real-world case study

Haribo's ERP Implementation Failure (2018)

- Enterprise Resource Planning (ERP) launched across 16 factories in 10 countries
- Poor process mapping & no infrastructure stress testing
- Inventory tracking failed → empty shelves
- Sales dropped 25% after rollout
- **Root causes:** underplanning, no load balancing, no monitoring
- Highlights need for capacity planning & observability



This case shows that even a beloved global brand can suffer major losses when on-premise systems are poorly managed.



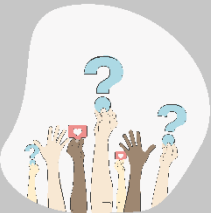
*How Haribo's ERP failure cost them millions,
Image source: [LinkedIn](#)*

e-learning recap

Reflecting on your learning...

The e-learning for this topic, covered the following areas:

1. Understanding On-Premise Infrastructure
2. Setting Up and Maintaining Infrastructure
3. Managing Computational Resources
4. Optimising Pipeline Efficiency
5. Load Balancing and Scheduling
6. Using Docker and Kubernetes On-Prem



- Do you have any questions about any of these areas?
- Did everything in the e-learning make sense?



Q&A discussion

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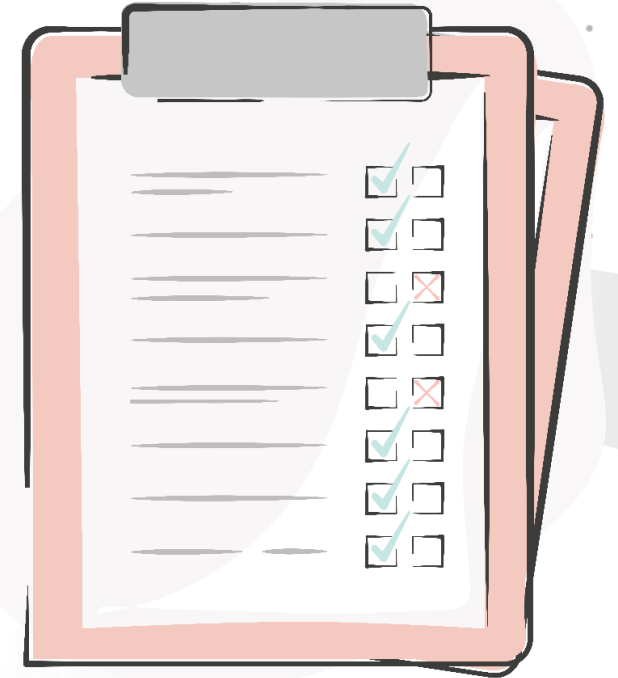
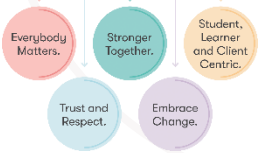


Webinar Agenda

Today, we will cover the following:

1. Core concept recap
2. Practical lab
3. Summary
4. Q&A

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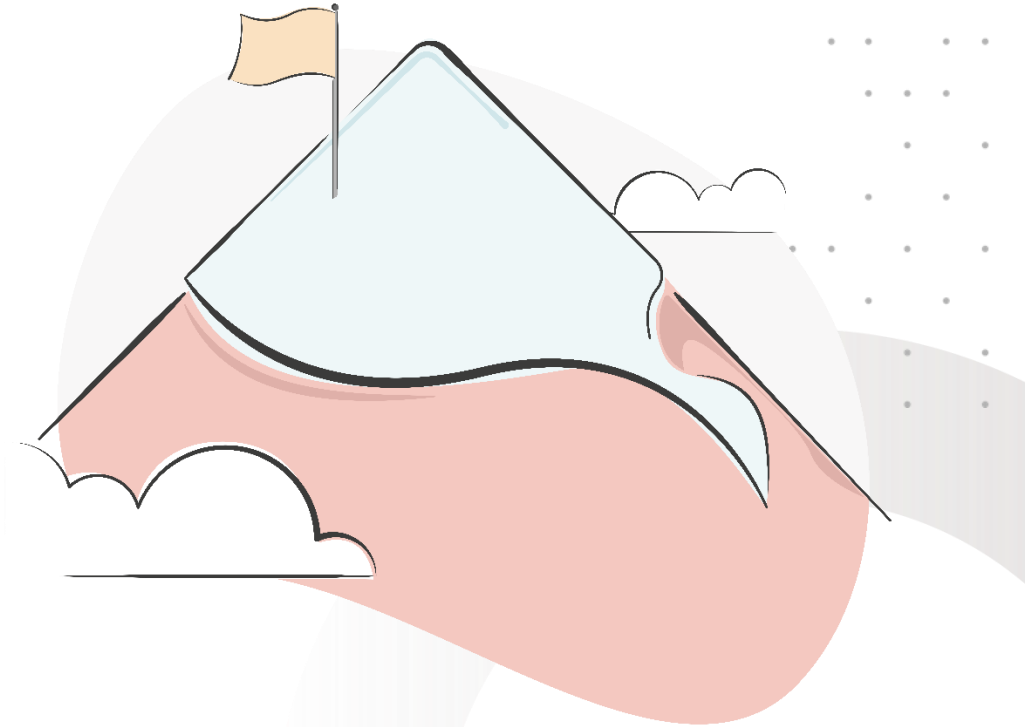
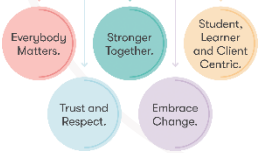


Session aim and objectives

By the end of this session, you should be able to:

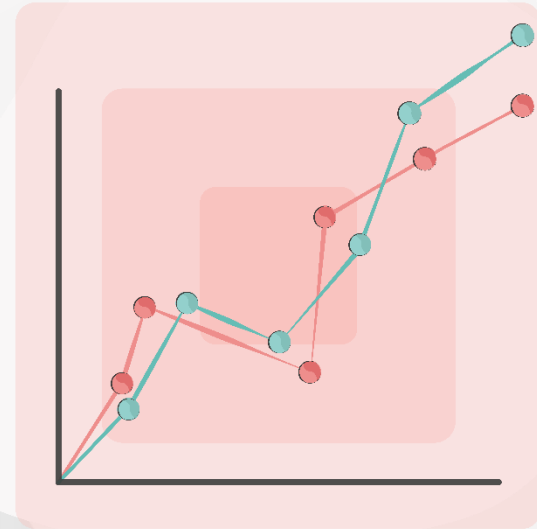
- Design and Maintain On-Premise Data Infrastructure
- Optimise Resource Usage for Data Pipelines
- Implement Scalable and Reliable On-Prem Workflows Using Modern Tooling

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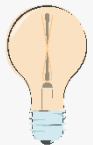
Recap of core concepts



Defining on-premise infrastructure

What Is On - Premise Infrastructure?

- Physical servers, storage, and networking
- Self - managed environments
- No auto - scaling or managed services

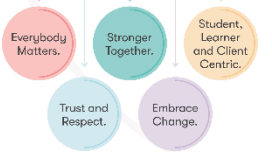


Unlike the cloud, where services are abstracted and scalable, on - prem systems require manual setup, configuration, and maintenance.



Figure: On-premise architecture, Source: [Datapac](#)

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Setup essentials

Building your infrastructure

- Plan for compute, storage, and redundancy
- Design secure network topology
- Harden OS and runtime environments

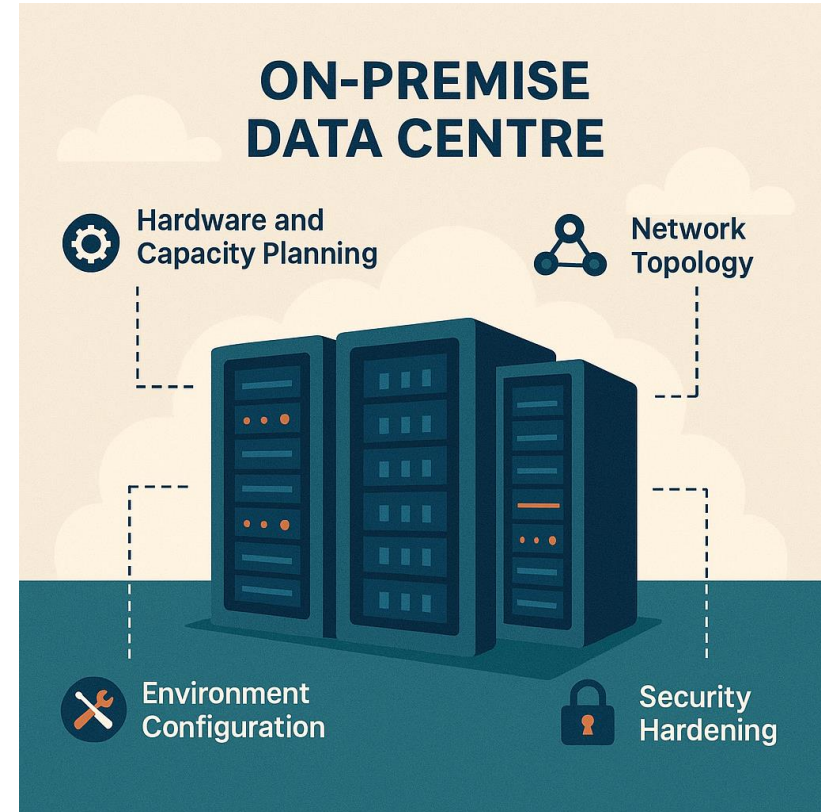
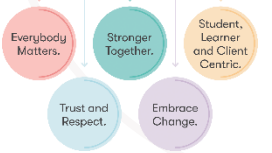


Figure: Key Considerations for Setting Up an On-Premise Data Centre

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Maintenance matters

Keeping systems healthy

- Patch regularly and automate updates
- Monitor CPU, memory, and disk
- Test backups and document everything

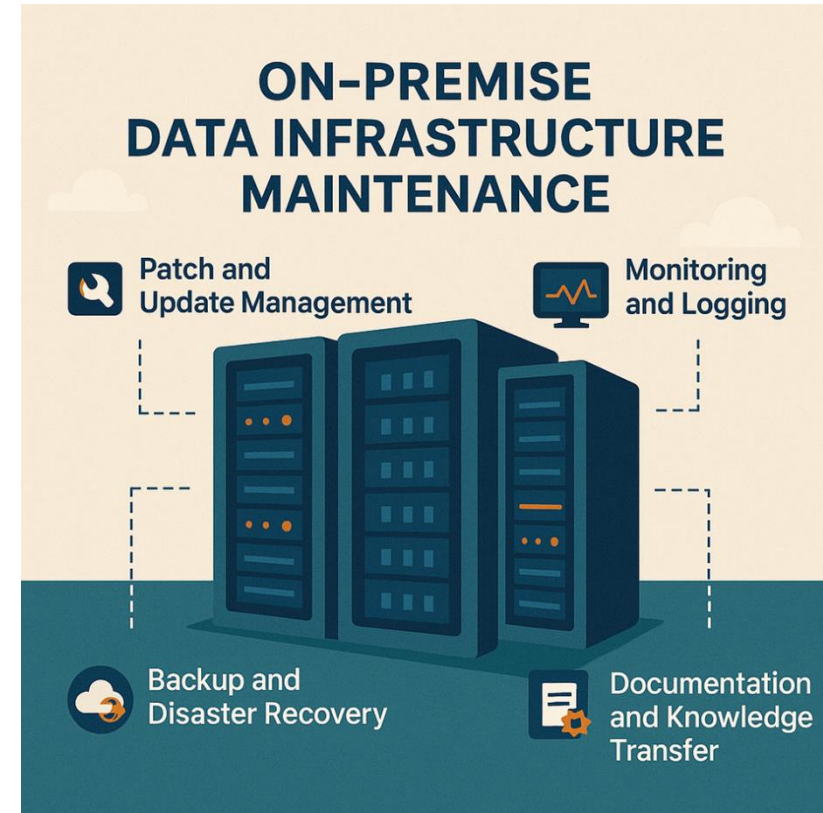
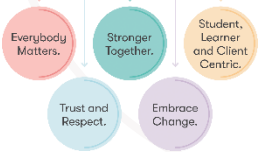


Figure: Ensuring System Health through Regular Maintenance and Monitoring

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Resource awareness

Understanding system limits

- CPU, RAM, I/O, and network are finite
- No “scale up” button
- Poor planning = system crashes

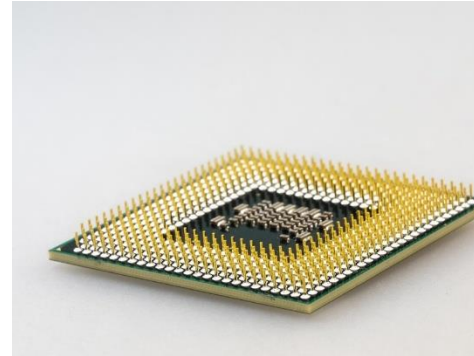


Figure: a Central Processing Unit (CPU)



Figure: a RAM stick, or memory module.

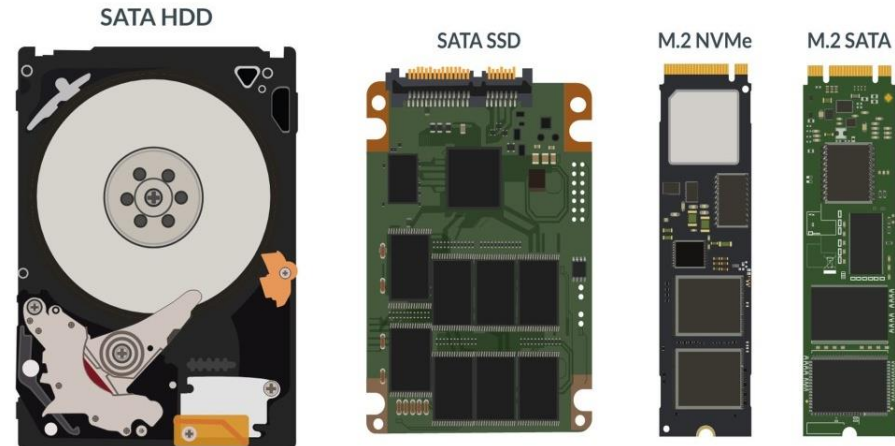
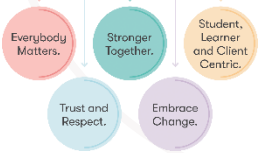


Figure: different types of hard disk storage.

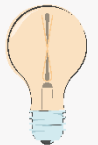
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Smart usage

Optimising resource consumption

- Chunk large datasets
- Use vectorised operations
- Avoid unnecessary I/O



These techniques reduce load on CPU, memory, and disk - keeping your system responsive and stable.

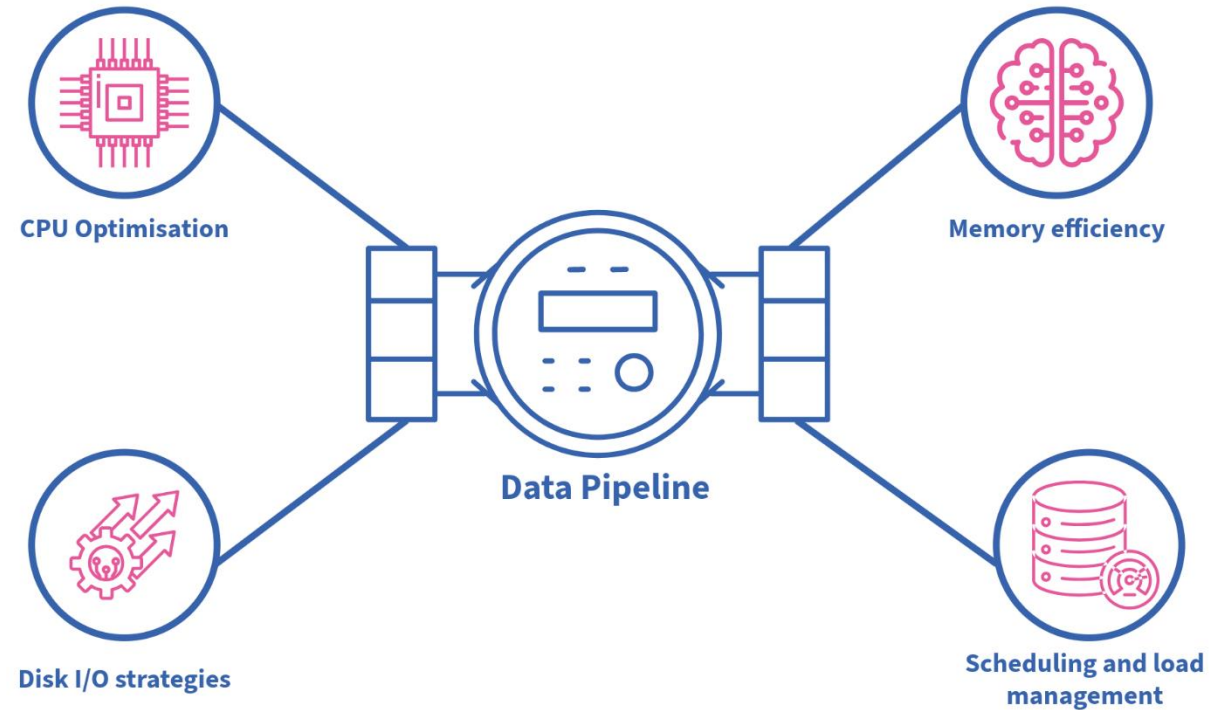
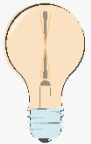


Figure: Ensuring System Health through Regular Maintenance and Monitoring

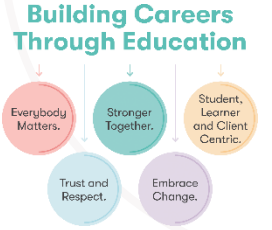
Scheduling strategy

Avoiding the midnight crash

- Stagger job start times
- Prioritise critical workloads
- Schedule off - peak execution



Tip: Don't schedule all jobs at once—stagger them, prioritise critical tasks, and use off-peak hours to keep systems running smoothly.

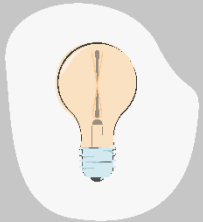


Like a shared kitchen, your infrastructure needs coordination - stagger jobs, prioritise tasks, and schedule smartly to avoid a system traffic jam.

Tooling up

Docker & Kubernetes On-Prem

- Manual provisioning and networking
- Local storage and registries
- Monitor everything yourself



Docker and Kubernetes are powerful
- but on - prem, you're responsible
for the plumbing.



Logos for two tools common to a on-premise
data infrastructure: Docker and Kubernetes

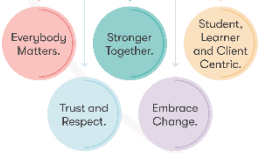
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Real - World Lessons

Case Studies in Action...

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Manufacturing: Staggered jobs
→ no more failures,

Image source: [itn.co.uk](https://www.itn.co.uk)



Hospital: Rehearsed
updates → enabled
rolling upgrades.

Image source:
[Tagmedstaffing.com](https://www.tagmedstaffing.com)

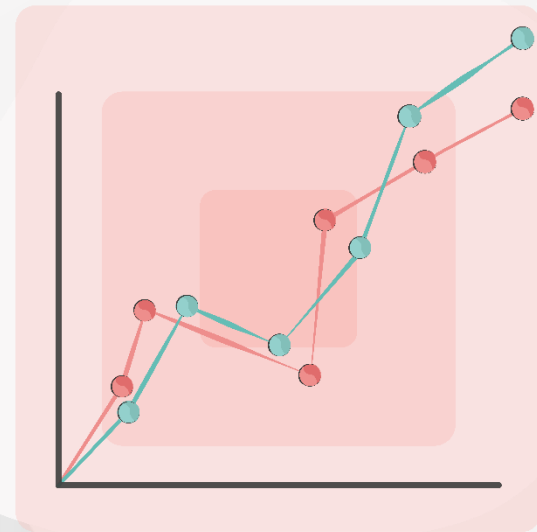


University: Added quotas
→ fairer GPU usage.

Image source:
[Universitiesac.co.uk](https://www.universitiesac.co.uk)



Practical lab



Exercise part 3

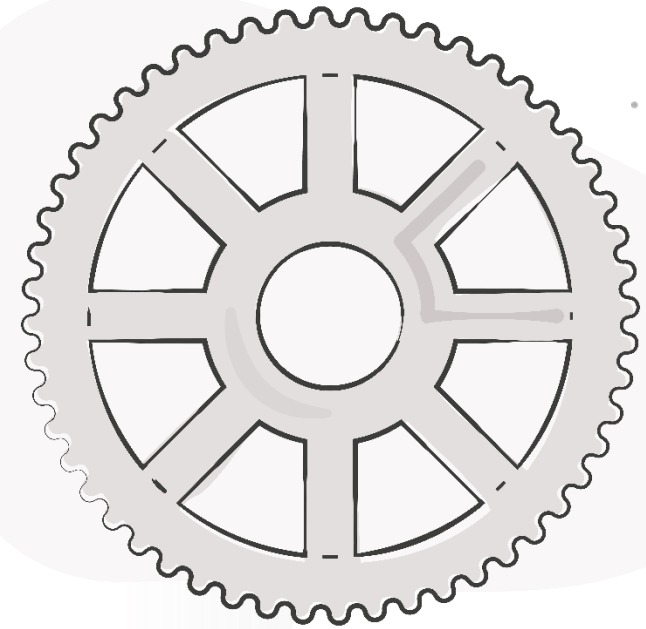
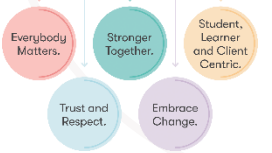
Next phase of design based on US data

- **Update Database Schema:** Modify the schema to accommodate data from the American version of the software application.
- **Re-implement Data Pipeline:** Develop a pipeline to clean, validate, and load UK, French, and American data into the revised schema.
- **Use Test Data:** Utilise provided test data for 10 American users and their login timestamps.
- **Future Phases:** Prepare to integrate data from additional countries with varying formats.
- **Documentation:** Establish comprehensive documentation for the schema and pipeline for future implementation by other teams.

Files Provided in the Hub:

- US User Data.csv: Contains 10 sample records.
- US-User-LoginTS.csv: Contains login timestamps for January 2025.

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Practical challenge

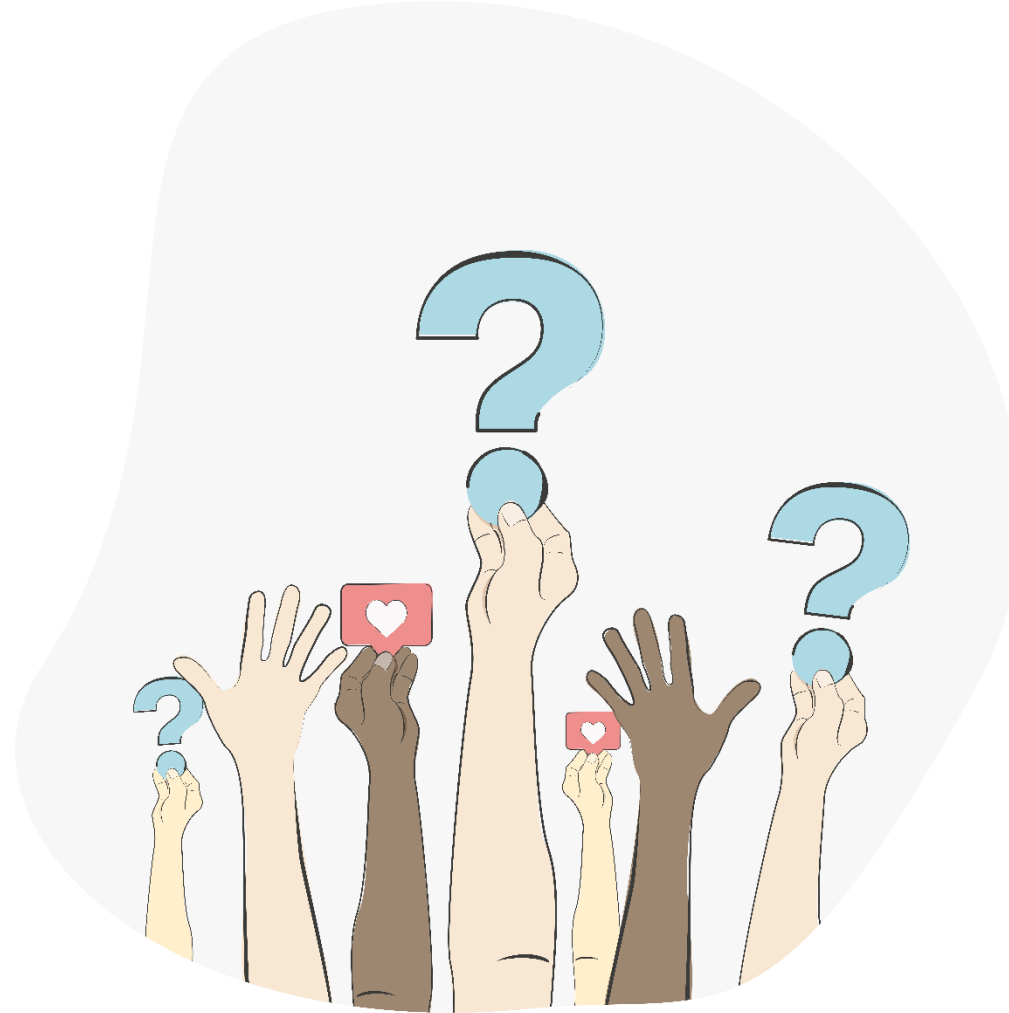


Key Learning Summary

- Running your own servers means owning everything from cabling to backups - and learning to think like a systems engineer.
- Know what happens under the hood when pipelines run - and why unmanaged resource use can bring everything else crashing down.
- Small adjustments like chunking data, limiting threads, or compressing files can make a big difference on tight on-prem systems.
- **Timing really matters** - when and how you run your jobs can prevent bottlenecks and keep shared systems running smoothly.
- Docker and Kubernetes still work on-prem, but you have to build the cloud-like environment around them yourself.
- Real-world success comes from visibility, good habits, and learning from the things that go wrong before they break everything.



Any questions or feedback?



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Thank you

